



BOILEAU GUILLAUME

Systems Engineer | IVV, Test Benches, V-Cycle, Agile & Software/Hardware Integration

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EXPERIENCE

Software Engineer - System Integration & Hardware Interfaces

VEV Platform Services France

📅 2025 - 2026

📍 Valbonne, France

Context: Development and integration of software services for EV charging infrastructure within a CPMS platform connected to operational hardware systems.

Objective: Support robust software integration, operational data flows and interface reliability between platform services and charging station systems.

- **Software development:** Developed web and backend services using TypeScript, Angular and Node.js in an operational industrial environment.
- **Agile environment:** Worked within an Agile delivery context, contributing to task prioritisation, iterative development and coordination with technical stakeholders.
- **Agile tooling:** Used Zenhub to support task tracking, backlog follow-up, iterative development and technical coordination.
- **System integration:** Contributed to the integration and maintenance of software services interacting with field hardware and charging infrastructure.
- **Hardware/software interfaces:** Analysed interactions between platform services, operational data flows and charging station systems.
- **Operational validation:** Checked data consistency, service behaviour and interface continuity in production-like conditions.
- **Incident analysis:** Investigated production issues involving software behaviour, data exchanges and hardware-connected systems.
- **Data exchange:** Implemented and monitored FTP-based operational data exchange services.

Senior R&D Engineer - Space Systems, IVV & Validation Workflows

Laboratoire Lagrange, Observatoire de la Côte d'Azur, CNES

📅 2023 - 2025

📍 Nice, France

Context: Engineering activities for the LISA ESA/NASA space mission, involving complex software chains, model integration, simulation workflows, validation campaigns and international coordination.

Objective: Develop, integrate and validate software and simulation workflows under strong consistency, traceability, performance and verification constraints.

- **Space systems:** Contributed to mission-level analysis workflows for the LISA ESA/NASA space mission in a large international technical collaboration.
- **V-cycle logic:** Supported engineering activities aligned with specification, integration, verification and validation logic for complex mission-level software workflows.
- **IVV logic:** Structured integration, verification and validation workflows for complex models, software chains and simulation outputs.
- **Test strategy:** Defined comparison campaigns, validation checks and acceptance logic for technical outputs produced by multiple contributors.
- **Simulation environment:** Developed and used numerical simulation workflows to assess system-level behaviours and performance limitations.
- **Anomaly analysis:** Investigated inconsistencies, deviations and unexpected behaviours affecting result reliability and interpretation.
- **Software platform:** Developed CATpyLISA, a Python platform for large-scale model integration, comparison, validation and technical review. [GitLab:CATpyLISA](#)
- **Technical coordination:** Coordinated contributors, supervised interns and supported validation activities across multiple stakeholders.
- **Technical documentation:** Used Confluence to support technical documentation, coordination notes, validation follow-up and knowledge sharing within project activities.

Data Scientist - Diagnostics, Test Analysis & Performance Validation

Universiteit Antwerpen, Belgium

📅 2021 - 2023

📍 Antwerp, Belgium

Context: Data analysis and performance studies for precision instruments in a high-requirement technical environment linked to gravitational-wave detector performance.

Objective: Identify, characterize and mitigate sources affecting system sensitivity using robust statistical, computational and validation-oriented methods.

- **Performance validation:** Analysed sources affecting system sensitivity, performance stability and measurement quality.
- **Technical diagnostics:** Identified contributors to performance degradation and supported prioritisation of mitigation actions.
- **Analysis pipelines:** Developed Python-based pipelines for noise source identification, characterization and mitigation.
- **Test data analysis:** Analysed measurement and simulation outputs to identify abnormal behaviours and possible degradation sources.
- **Validation reporting:** Produced reproducible and traceable technical analyses for international engineering and research stakeholders.
- **System impact:** Contributed to studies identifying environmental and instrumental constraints for next-generation gravitational-wave observatories.

PROFILE FOR THALES

Systems engineer with a strong background in space systems, software/system IVV, simulation workflows, test analysis, anomaly investigation and hardware/software interface understanding. Experience developed through major international collaborations including LISA, LIGO-Virgo-KAGRA and Einstein Telescope, complemented by recent industrial software experience involving operational services connected to hardware infrastructure.

For an IVV / test bench / avionics and system integration role, I bring a practical engineering mindset: working within V-cycle and Agile delivery contexts, defining validation logic, integrating complex software chains, analysing test and simulation outputs, investigating anomalies, documenting technical evidence and coordinating with multidisciplinary stakeholders.

- Strong experience with complex systems under integration, validation, consistency and traceability constraints.
- Verification and validation of software chains, simulation workflows, model outputs and technical deliverables.
- Analysis of anomalies, interface behaviours, performance degradation and system-level limitations.
- Experience with software/hardware interfaces through operational industrial systems connected to charging infrastructure.
- Ability to interface with technical contributors, engineers, researchers, project stakeholders and international teams.
- Structured documentation and technical reporting in English for high-standard collaborative environments.
- Understanding of V-cycle engineering logic: requirements, specification, integration, verification, validation and technical evidence.
- Experience working in Agile software environments with iterative development, task coordination and stakeholder interaction.
- Use of collaborative engineering tools including Confluence for technical documentation and Zenhub for Agile task tracking and coordination.

LANGUAGES

English



Spanish



EDUCATION

PhD in Physics

Université Côte d'Azur

📅 2018 - 2021

📍 Nice, France

Selective Graduate Program in Fundamental Physics (Magistère)

Université Paris-Saclay

📅 2016 - 2018

📍 Orsay, France

MSc in Astronomy and Astrophysics

Observatoire de Paris / Université Paris-Saclay

📅 2017 - 2018

📍 Paris, France

Junior R&D Engineer - Signal Analysis, Simulation & Software Validation

ARTEMIS, Observatoire de la Côte d'Azur

2018 – 2021 Nice, France

Context: Engineering and analysis activities for gravitational-wave mission preparation, involving signal analysis, simulation, software workflows and noise characterization.
Objective: Develop robust analysis methods for complex signals and support technical investigations in demanding system environments.

- Signal analysis:** Developed methods for the separation of overlapping low-SNR signals in complex datasets.
- Simulation workflows:** Built simulation approaches for complex signal environments, uncertainty studies and large-scale datasets.
- Software validation:** Compared simulated outputs, model behaviours and analysis results to assess consistency and robustness.
- Technical investigations:** Investigated noise sources through cross-sensor correlation analyses in diagnostic contexts.
- Mission performance:** Supported the identification of limiting factors affecting mission-level performance and system sensitivity.

PROJECTS

Co-author and full member of the LISA Consortium

ESA/NASA Space Mission - Large-scale International Collaboration

2018 – now

Contribution to software, analysis and validation workflows for the LISA space mission within an international engineering and scientific collaboration involving ESA, NASA and multiple research institutes.
Role: Software workflows, technical coordination, model integration, simulation campaigns, validation activities, complex signal-processing tasks and collaborative engineering support.
Relevance for IVV / test benches: Work involving complex system behaviour, mission-level performance constraints, simulation-based validation, consistency checks, technical documentation, anomaly investigation and cross-functional engineering coordination.
Program scale: Major international space mission with an estimated budget of approximately 2 billion EUR over 10 years.

Co-author and full member of Einstein Telescope and LIGO–Virgo–KAGRA Collaborations

International Collaborations

2021 – now

Contribution to collaborative developments in data analysis, signal characterization, software methods and technical investigations within the LIGO–Virgo–KAGRA and Einstein Telescope collaborations.
Role: Development of analysis tools, contribution to technical activities and support to complex software workflows in high-standard collaborative environments.
System impact: Studies on environmental and instrumental noise sources helping identify fundamental design constraints and performance limitations for next-generation gravitational-wave observatories.

STRENGTHS

IVV, Test Benches & Systems

- IVV
- IVVQ
- V-Cycle
- Agile
- Requirements
- Specification
- Acceptance Criteria
- System Integration
- Software Integration
- Test Strategy
- Test Plans
- Validation Campaigns
- Simulation-Based Validation
- SIL Logic
- HIL Understanding
- System-Level Analysis
- Space Systems
- Mission-Level Performance
- Requirements Verification
- Traceability

Diagnostics, Interfaces & Test Analysis

- Anomaly Analysis
- Root Cause Analysis
- Technical Diagnostics
- Performance Analysis
- Test Data Analysis
- Interface Analysis
- Hardware / Software Interfaces
- Operational Reliability
- Data Consistency
- Signal Processing
- Noise Analysis
- Uncertainty Analysis
- Technical Reporting

Tools, Software & Coordination

- Python
- Linux
- Bash
- Git
- GitLab
- GitHub
- Docker
- CI/CD
- SQL
- TypeScript
- Angular
- Node.js
- REST API
- FTP
- Confluence
- Zenhub
- Agile Tooling
- Technical Documentation
- Technical Coordination
- Stakeholder Communication
- Technical English

BSc in Fundamental Physics

Université Paris-Saclay

2015 – 2016 Orsay, France

PROFESSIONAL TRAINING

Supervised Machine Learning (Regression & Classification)

DeepLearning.AI - Andrew Ng

2020 Coursera